

Exploring Human Behaviours of Stress through Virtual Reality Interactions using Accessible, Embedded Sensors and User-Personalised Machine Learning

Saskia Davies
Computer Science
Swansea University
Swansea, United Kingdom
saskia.davies@swansea.ac.uk

Tom Owen
Computer Science
Swansea University
Swansea, United Kingdom
t.owen@swansea.ac.uk

Sean Peter Walton
Computer Science
Swansea University
Swansea, United Kingdom
s.p.walton@swansea.ac.uk

Abstract

Virtual reality (VR) is increasingly applied in therapy due to its immersive and adaptable qualities, often augmented with sensors to facilitate the behavioural analysis of users. However, research in this domain commonly employs complex or intrusive setups that hinder real-world impact, arising from practicality limitations. In response, we aim to explore the capability of standalone VR sensors in stress detection. We gather a physiological dataset through VR user studies with 41 participants. A personalised random forest model is created for each user, trained against a ‘stress index’ produced by a medical-grade sensor. These models achieve 83.77% accuracy on average, and feature importance analysis identifies hand behaviours to be most significant for stress prediction. Further preliminary analysis suggests that high-velocity hand movements may be indicative of high stress levels. Results demonstrate how VR stress prediction can be achieved without complex setups, while informing the modalities and behaviour analysis of future systems.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Computing methodologies** → **Supervised learning by classification**; • **Applied computing** → **Psychology**.

Keywords

virtual reality, affective computing, multimodal sensors, personalised machine learning, cognitive behavioural therapy, exposure therapy

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1 Introduction

Mental health is thought to be one of the most promising applications of virtual reality (VR) in healthcare [30, 31, 33]. In recent years, VR has been progressively developed for cognitive behavioural therapy (CBT) and exposure therapy (ET) due to its immersive qualities, shown to elicit more naturalistic user responses than traditional laboratory sessions [28]. Supplementing these approaches with multimodal sensors further enables researchers and clinicians to understand the cognitive and emotional responses of patients throughout VR therapy and adapt therapeutic stimuli accordingly.

Human-centred design is a vital integration to understand patients and their interactions during virtual therapy. However, we find an imbalance between the usability and practicality of sensor-augmented VR systems throughout existing research. To achieve the best results, state-of-the-art works often use intrusive, high-end sensors [16, 18], or combine a complex array of separate external devices [16]. While this limits reproducibility and accessibility [16, 18], the real-world application of these systems and their tangible impact also suffers as a result.

In response, we explore (1) whether standalone VR sensors can estimate user stress – a common measurement in virtual CBT – comparable to a medical-grade sensor, and (2) which sensing modalities and resulting human behaviours contribute most towards stress detection in VR. We gather an expansive dataset of 41 participants’ physiological data during a range of VR experiences, aiming to define the way in which people react and interact in VR environments. Using these sensors may inform a more usable and accessible design for future applications based on real user data.

2 Related work

The growing demand for access to therapy has triggered large-scale developments in eHealth innovation [3, 23], which frequently employs VR technology. While escapism and distraction are common applications of virtual therapy [18], it is predominantly applied to methods of ET, in which patients are subjected to virtual stimuli representing their fears or anxieties. These approaches provide an opportunity to disconfirm negative expectations and practice positive behavioural responses in return [29, 31].

Much of the existing research surrounding VR therapy comprises correlating environments when targeting certain affective states. For example, phobia-related ET commonly adopts situations surrounding public speaking [22], high places [17] or confined spaces [34]. Environments designed to induce stress often focus on unpredictable or fluctuating scenarios such as rollercoaster rides [19],

while relaxation therapy typically includes naturistic forms of escapism, placing the user in a forest [9], beach [2], or underwater setting [1]. Though these environments may be interactive, desired results can also be triggered by passive VR video experiences [5].

However, immersive technology researchers highlight that, despite the precise control supported by VR regarding a user's external experience, insight into their internal experience remains unknown due to a lack of understanding of their physiological and emotional responses [16]. As such, there is scope to integrate an affective understanding of users through multimodal sensors, not only to understand their underlying psychology, but also to influence the VR environment. Such analysis provides a means of recognising the behaviour emerging from underlying physiological readings in response to a VR experience [11, 14].

Various sensor modalities are adopted across affective computing literature. The face is argued to be one of the most valuable indicators of generalised affect, along with cardiovascular data, eye tracking features, skin conductance and movement data, such as head and hand motions [16]. Given the importance of psychological fear activation during treatment [13, 31], anxiety and stress detection are frequently employed in ET interventions. Heart rate, heart rate variability (HRV), skin conductance (EDA) and respiration rate are often utilised to identify anxiety levels [18], and while electroencephalogram (EEG) measures provide highly accurate results [4, 10, 35], a simple wearable sensor integrating blood volume (PPG), EDA and skin temperature readings has been shown to produce comparable results [8]. In VR environments specifically, HRV and EDA are the most common sensors used to measure anxiety [12].

Similarly, stress can also be characterised by cardiovascular activity, EDA, skin temperature, respiration and motion activity [18, 32]. However, HRV is stressed to be one of the most influential indicators of stress [21, 27], while the importance of PPG and EDA in stress detection has been demonstrated in VR settings [8]. On the other hand, relaxation is a common counter-measurement to assess the effectiveness of treatment, in which most research focusses on HRV and respiration [20]. Accelerometers can provide further information about levels of focus during meditative states, aiding relaxation recognition [20].

The increasing availability and practicality of sensors is prompting physiological data use in VR systems at an exponential rate [25]. Many VR headsets now come with various sensors, such as for eye tracking and head movement, already embedded for behavioural analysis [18]. Despite aiming for accessibility, some argue that such systems currently lack practicality outside of laboratory environments. A key reason concerns the complexity of sensors and their setup, as many state-of-the-art works rely on multiple external and separate devices to be placed on the user [16], particularly when trying to use low-cost modalities [15]. For example, complex EEG equipment can involve more than 30 individual electrodes [35], while some motion captures require sensors to be placed on every part of the body [26]. This can cause issues with sensor synchronisation and setup, limit reproducibility, and particularly from a deployment point of view, hinder feasibility and utility [16]. However, even though sophisticated sensors can generate good results in isolation, they are typically more intrusive [18] and difficult to combine with VR devices [16] given the lack of access to head and face

features while wearing a headset. Alongside expense and resulting deployment limitations, this introduces VR-specific complications as intrusive sensor setups often provide the most accurate results [18].

Managing the balance between usability, practicality and accuracy is therefore a critical task for designing therapeutic VR systems. To achieve real-world implementation, such devices must surpass the research phase in order to reach full potential in their therapeutic purpose and tangible impact. We argue that more user-centric design is needed, harnessing real user data captured in VR to prioritise usability. Generating high quality data in this domain is essential to gather tailored user insights and inform the design of future therapeutic systems.

3 Virtual reality and data capture design

In response to the real-world limitations of existing work, we designed a VR prototype that, through data collection, facilitates the investigation of stress detection using only sensors that are currently embedded in commercial VR devices. Our primary goals are to assess this feasibility, and to define which sensing modalities are best suited to VR-specific stress detection. We achieved this by compiling a dataset of participants' physiological reactions in response to various VR scenarios, providing a range of user data directly in response to VR experiences.

We used a Meta Quest Pro VR headset for this study, chosen for its native eye and face tracking capabilities which therefore eliminate the need for external sensors. As with typical VR devices, embedded accelerometers and gyroscopes also allow for tracking of head and hand movements through the headset and controllers. While these sensors shaped the core of our training data, our data labels are determined using a wireless eVu-TPS finger-clip sensor, with tracking capabilities for HRV, EDA, PPG and skin temperature built in. This medical-grade sensor calculates a 'stress index' value using proprietary algorithms from Thought Technology Ltd., forming our target variables.

Meta Quest Pro	eVu-TPS
Headset tracking (3D position, 4D rotation)	Heart rate variability (HRV) (ms)
Controller tracking (3D position, 4D rotation: left + right)	Skin conductance (EDA) (μ S)
Facial feature tracking (40 facial features: left + right)	Blood volume (PPG) (SpO2)
Eye tracking (3D position, 4D rotation, 3D gaze: left + right)	Skin temperature (°C)

Table 1: An overview of physiological features incorporated in our dataset, captured from sensors in the Meta Quest Pro and eVu-TPS.

To populate our dataset, we incorporated a variety of virtual experiences (Figure 1) into a VR application, aiming to evoke a broad spectrum of physiological responses. Given the scientific justification of video usage in VR [5], we obtained a superset of 10 publicly

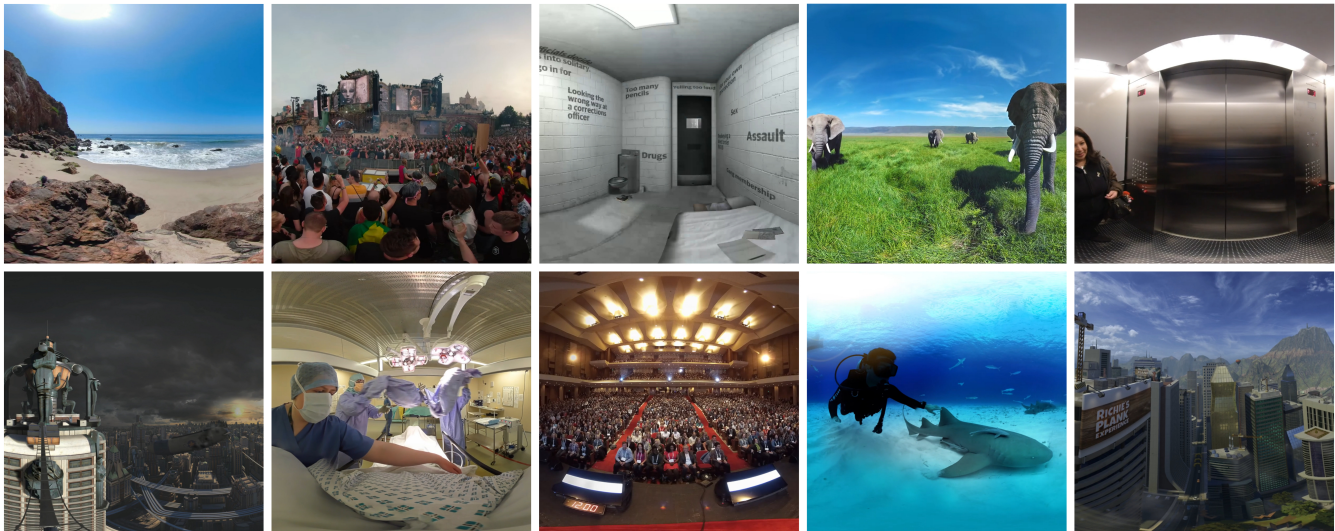


Figure 1: Screenshot of each 360-degree video experience included in the VR application. Chronologically, the images shown include scenarios for: beach relaxation, music festivals, educational prison experiences, wild animal encounters, elevators, high places, hospital scenarios, public speaking, underwater excursions and free-falling.

available 360-degree videos which, in line with previous work, are each designed for a specific emotional experience. All videos were trimmed to a uniform length of 1:20 minutes to standardise data collection across all experiences.

The final system presented a randomised subset of six videos in sequence, ensuring a balanced and manageable experience for participants. First, an instructive splash screen is displayed during sensor calibration, and then sensor data is captured passively at a fixed frequency of 50Hz throughout video playback. We also included a skip function that allows users to bypass any video that causes discomfort, with clear controller instructions explained on the initial splash screen. Instructions for removing the headset and finger clip sensor appear upon completion of the session.

4 User study and data collection

Participants of this study were recruited primarily through university advertising. We gathered a total of 41 participants (28 male, 12 female, 1 non-binary; 13 aged 18–24, 22 aged 25–34, 6 aged 35+) with varying levels of VR experience and relatability to common phobias.

The studies were carried out as one-to-one sessions, lasting approximately 20 minutes in total including the initial completion of a demographic questionnaire. Following a verbal study briefing, participants were equipped with the VR headset and eVu-TPS sensor. While these sensors passively capture physiological data, participants engaged with the six immersive videos allocated by our application. For safety and comfort, participants were seated in a swivel chair throughout the VR session, preventing them from inadvertently moving into obstacles, such as the room's walls, while still enabling a panoramic view of the virtual environment.

Upon completion and removal of the headset and sensor, a post-study interview comprised a series of questions regarding their

overall experience and emotional responses. All resources and procedures were approved by the Research and Ethics Committee at Swansea University.

5 Data analysis methodology

The scope of this research is to perform an introductory analysis on the quantitative physiological data gathered from users during each study. We employed a number of pre-processing steps on the data to assist with evaluation and machine learning analysis. Aiming to



Figure 2: Example image of the system setup during user studies. The participant is sat on a swivel chair in a spacious, isolated room, and the Meta Quest Pro is strapped to their head while they hold the VR controllers. The eVu-TPS sensor is attached to their right-hand index finger.

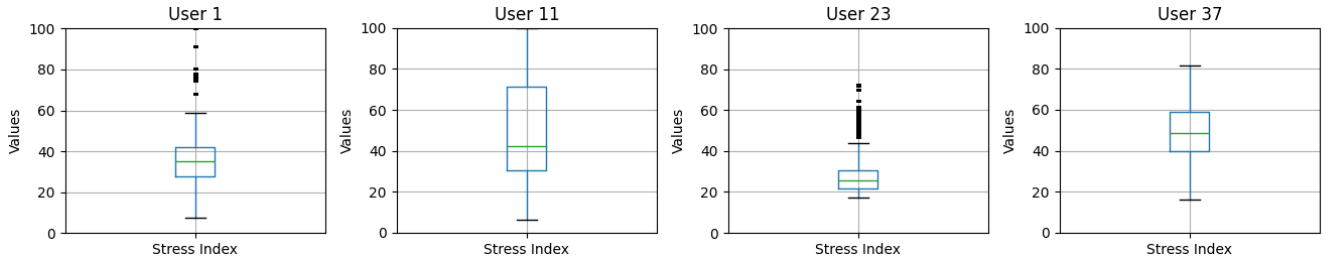


Figure 3: Box plots denoting stress index variance of users 1, 11, 23 and 37 over the course of their respective VR sessions.

reduce the high dimensionality of our data, we first removed any unchanging (eg, tongue facial features) or non-comportmental (eg, sensor contact) features in the dataset. On the remaining data, we performed feature engineering to combine composite features: bilateral facial components are averaged into a single value, while 3D movement and 4D rotation are each combined into a single velocity value. These methods condensed the original 104 dimensions into 41.

Secondly, to improve interpretation for categories of stress, we split the continuous stress index into three discrete classes: low, medium and high stress. As shown in Figure 3, the distribution of stress levels varies broadly between users, and so we divided each user’s stress index into thirds based on proportions of their individual training sets. This ensures that the labels for each model reflect the user’s personal low, medium and high stress levels, as these boundaries are clearly not universal.

Using discrete stress index as target variables, a personalised model was then trained for each individual user. We used a random forest classifier [6] - an ensemble of tree classifiers that is fast and resistant to noise [7] - with random search optimisation and cross-validation in order to maximise both performance and generalisation. Given the diverse structures of our features, this tree-based model is especially beneficial due to its inherent handling of non-linear relationships, without the need for scaling or standardisation [7].

To interpret the models’ decision-making processes, SHapely Additive exPlanations (SHAP) [24] was implemented to identify the most important features – and therefore the most important

sensor modalities – in understanding user stress levels in VR. SHAP provides a transparent method to quantify the importance of features and how they contribute to the model’s output [24]. As SHAP values are generated for each data point, we preliminarily explore the top-ranking physiological characteristics of one user, visualising this data to uncover specific behaviours that influence stress detection.

6 Results

Upon training each personalised model, we achieved an average accuracy of 83.77%. This is comparable to that produced by other modalities in the field, which average at around 80% and rise towards 98% [18]. An example of the highest, median and lowest accuracies is shown in Figure 4, where classes 0, 1 and 2 represent low, medium and high stress respectively.

Given the conclusive accuracies of these models, we assume its feature importance calculations to be sufficiently applicable to human behaviours of stress. While there is some variation in feature importance rankings for each user, hand velocity behaviours are among the top contributors for all user models, followed by head velocity. By averaging the overall importance of each feature across all models (Figure 5), controller and headset velocities are shown to be most globally indicative of VR-based stress detection across the participant population. Facial features are harnessed much less, although a variety of regions participate to model outputs with similar levels of general importance.

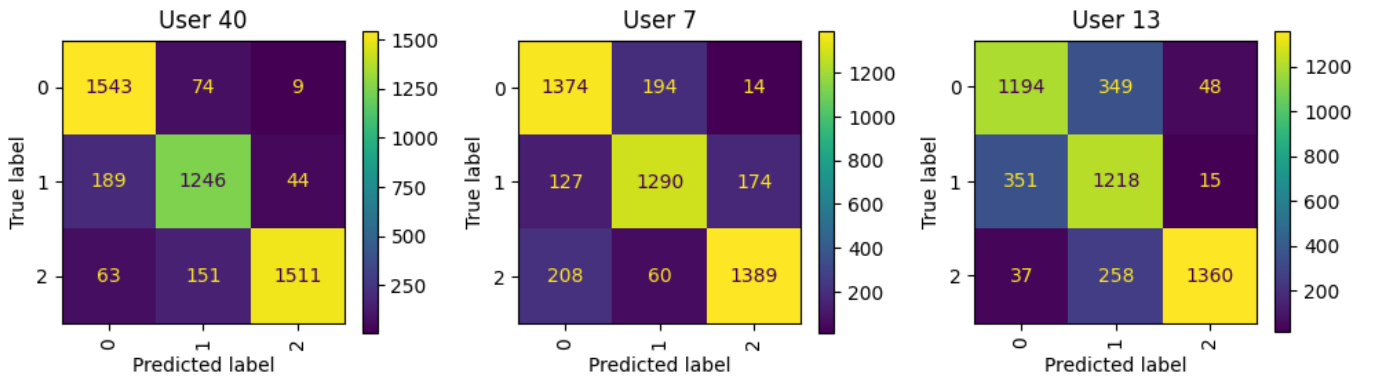


Figure 4: Confusion matrices for models with the highest (user 40, 89.03%), median (user 7, 83.91%) and lowest (user 13, 78.1%) accuracies. The the model’s predicted stress class (x-axis) is compared against the real class label (y-axis) from its testing dataset.

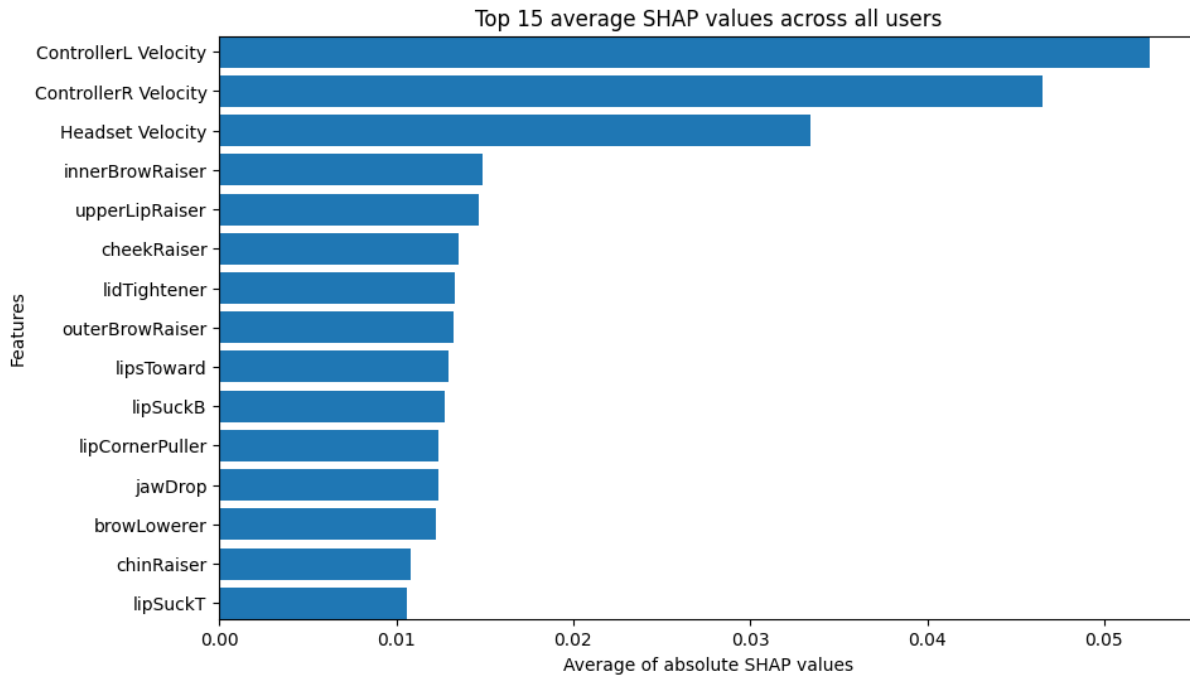


Figure 5: Bar chart of average SHAP values across all user models, showing the top 15 features that contribute most towards stress detection in VR.

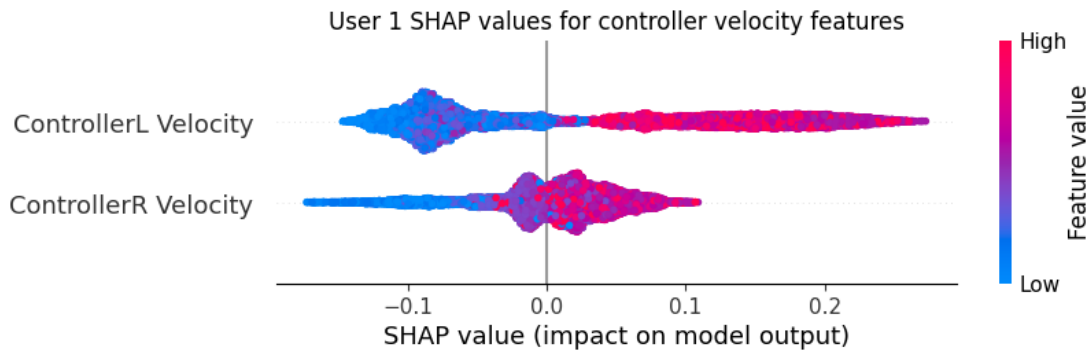


Figure 6: Beeswarm plot comparing raw values against SHAP values for left and right controller velocity, specifically those contributing towards high stress classification. Positive SHAP values indicate a higher influence on the model's prediction of high stress, while negative SHAP values indicate an opposing influence on high stress prediction.

To investigate how controller velocity contributes towards stress, we preliminarily graph the raw values of these features for User 1 against the SHAP values contributing towards 'high stress' classification. The beeswarm plot in Figure 6 shows a positive correlation between high velocity values and high SHAP values for both left and right controllers. In User 1's personalised model, this indicates that high-velocity hand movements contribute greatly towards the classification of high stress, further suggesting that rapid hand movement is an indicator of stress for this user. In contrast, low velocity values primarily correlate with a rejection of high stress classification.

We finally animate these movement behaviours using the original position and rotation values as seen in the videos linked below^{1,2}, where the controller colours become more red with higher SHAP values. Re-enacting User 1's study in this way, regions with high SHAP values appear to correlate with swaying and small wobbling motions, showing that these particular behaviours have a positive influence on high stress classification. As such, these mannerisms may be related to the high-velocity hand movements that are indicative of stress for User 1.

¹<https://youtu.be/40M5gbcSC7s>

²<https://youtu.be/kI2DyLfZWD4>

7 Discussion

Aiming to improve upon the real-world impracticality of existing research in the field, we worked towards the investigation of (1) whether standalone VR sensors are capable of measuring stress compared to other modalities, and (2) which sensing modalities and human behaviours contribute most towards stress detection. Our machine learning analysis shows that personalised models can predict user stress levels with an accuracy comparable to other modalities across affective VR literature, averaging at 83.77%. While these results prove the suitability of VR-embedded sensors, it also indicates that external sensors are not always necessary in understanding users. We believe that using standalone VR sensors can make virtual therapy systems more feasible and reproducible than using multiple external sensors or intrusive sensor devices, and therefore increase their everyday utility.

Secondly, our feature importance analysis demonstrates that VR controller velocities contribute towards stress prediction more than any other VR sensing modality. Controller velocity for one or both hands was among the top ranked features for every user model, and the global average SHAP values of controller velocities were each approximately three times larger than values of the highest-ranking facial features.

Through an introductory investigation of User 1's feature values, we discovered that high-velocity hand movements significantly correlate with this model's prediction of high stress. Moreover, supplementary observation links these results to swaying and small wobbling movements. While this may differ between users, these results imply that such behaviours are a result of incurred stress in VR environments, and may be indicative of stress for certain individuals. This correlates with previous work attributing "fast hand motions" to stress in VR [32], however we do not enforce specific interaction mechanisms, such as the swatting mechanic employed with VR controllers in this study. This may suggest that (1) our findings are instead representative of unrestricted, baseline VR behaviours, and (2) these results still hold for virtual environments that require controller interactions. More in-depth analysis is required to explore the integrity and generalisability of these insights, particularly accounting for the potential of uncontrolled variability incurred by not restraining hand movements.

Model generalisability is another aspect that requires additional research. While user-personalised models produce higher accuracy, the universal application of such methods is imperative, particularly for new users with little to no available data. These cold-start scenarios present a significant challenge given the heterogeneous data between individual users, however several strategies may account for this in real-world implementations. For example, demographic data may be leveraged as a means to compare similar users and therefore extract relevant existing data, or transfer learning techniques can adapt to new users by harnessing knowledge from pre-trained models. However, even after integrating these approaches, the model's early-stage performance may be less accurate until sufficient user-specific data is accumulated, highlighting the accuracy trade-off between personalisation and generalisability. Pre-session calibration may be beneficial in practice, granting a preliminary collection of tailored user data to be used by the model.

In practical therapeutic workflows, our contributions can be used towards valuable real-time feedback. This is beneficial not only to educate users and clinicians of internal affective experiences, but also to dynamically update the virtual environment in response to induced stress. For example, applications such as ET require a psychological fear activation from users for successful treatment, and particularly benefit from adapting feared stimuli according to user reactions.

As such, understanding the importance of controller behaviours in VR stress detection allows more informed development for future therapeutic VR systems. A particular emphasis should be placed on hand-based physiology, with a design that accommodates the analysis of these behaviours. Contrarily, we found eye tracking and facial features to be much less contributive overall. However, we hypothesise that such measures of emotion are more relevant in social contexts, where the face is not obscured and users can see their physical surroundings, rather than being immersed in a passive virtual world. This suggests that more sophisticated VR devices with in-built eye and face tracking may not be necessary for stress detection, however this data may be beneficial in other application contexts, such as multiplayer or agent-based social interaction in VR.

8 Contributions

This work poses several research contributions in the fields of HCI and therapeutic technology. Regarding technical contributions, we have proved the feasibility of standalone VR sensors in measuring stress, identified the importance of hand behaviours, and preliminarily explored what these specific behaviours might be – namely, rapid hand movements in swaying and wobbling motions (for User 1). To the general field of HCI, we have taken steps to improve usability and practicality in order to make VR therapy more accessible – therefore aiming to increase its real-world impact for people who benefit from treatment – and quantified the interaction behaviours of users to inform the design of future applications.

9 Conclusion

To summarise, we developed a VR application that passively records physiological data while users engage with a variety of virtual experiences, obtaining a valuable dataset through user study data collection. Using only VR-embedded sensors, personalised random forest models demonstrated an average accuracy of 83.77% in predicting stress based on user interactions and responses in VR. SHAP analysis revealed that controller movements are the most globally influential features in stress prediction, and a preliminary investigation into one user's controller behaviours suggests that rapid hand movements might be indicative of stress for certain individuals. Overall, results indicate that VR sensors can predict user stress levels in isolation, where future systems should focus on hand behaviours to achieve this.

Future work must explore in more depth regarding mannerisms that contribute to stress prediction, analysing the integrity and generalisability of these findings across different user cohorts. This includes the investigation of interactive environments that represent therapeutic contexts more realistically, including clinical studies directly with the target audience. Understanding responses

to stress in this way will be beneficial for the future development of therapeutic technology, and provide a better understanding of VR users universally.

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